

Amazon - 100 Technical MCQ Questions

Amazon (SDE-1, New Grad)

For BTech Computer Science / IT Students

Object-Oriented Programming

1. Which access modifier makes a member accessible only within the same class?

- A) public
- B) protected
- C) private
- D) default

2. An abstract class in Java can have:

- A) Only abstract methods
- B) Only concrete methods
- C) Both abstract and concrete methods
- D) No methods at all

3. Constructor overloading is possible when:

- A) Constructors have same name but different parameter lists
- B) Constructors are private
- C) A class has no constructor
- D) Only one constructor exists

4. What does 'this' keyword refer to in Java/C++?

- A) The parent class
- B) The current object instance
- C) A static variable
- D) The compiler

5. Which of the following supports multiple inheritance in Java?

- A) Classes
- B) Interfaces
- C) Constructors
- D) Packages

6. Which keyword is used in Java to prevent a class from being inherited?

- A) static
- B) final
- C) private
- D) abstract

7. What is encapsulation?

- A) Hiding implementation details and exposing only functionality
- B) Creating multiple classes
- C) Compiling code faster
- D) Using loops effectively

8. What is a virtual function used for in C++?

- A) To achieve runtime polymorphism
- B) To avoid inheritance
- C) To create static memory
- D) To improve compilation speed

9. Which of these best defines an interface in Java?

- A) A class with only static methods
- B) A contract of methods a class must implement
- C) A private class
- D) A type of exception

10. What is the main difference between an abstract class and an interface (pre-Java 8)?

- A) Interfaces can have constructors
- B) Abstract classes cannot have any methods
- C) Interfaces cannot have method bodies while abstract classes can
- D) There is no difference

DBMS & SQL

11. What does DBMS stand for?

- A) Database Management System
- B) Data Backup Management Service
- C) Database Model Structure
- D) Data Base Modeling Software

12. Which of the following best explains 'eventual consistency' in distributed/NoSQL databases?

- A) All nodes are always immediately consistent
- B) Given enough time without new updates, all nodes will converge to the same value
- C) Data is never consistent
- D) It refers only to backup consistency

13. In SQL, what does the LIKE operator do?

- A) Performs pattern matching using wildcards such as % and _
- B) Compares two numeric values
- C) Joins two tables
- D) Groups rows

14. What is a correlated subquery?

- A) A subquery that references a column from the outer query
- B) A subquery that never executes
- C) A subquery used only in INSERT statements
- D) A subquery without a SELECT clause

15. What is the main advantage of using an ORM (Object-Relational Mapping) tool?

- A) It eliminates the need for a database entirely
- B) It lets developers interact with the database using objects instead of raw SQL
- C) It automatically normalizes data
- D) It replaces indexes

16. What is a materialized view?

- A) A view whose result set is physically stored and periodically refreshed
- B) A temporary table only
- C) A synonym for an index
- D) A type of trigger

17. Which SQL command grants specific privileges to a user?

- A) REVOKE
- B) GRANT
- C) ALLOW
- D) PERMIT

18. What is the purpose of database normalization?

- A) To reduce data redundancy and improve data integrity
- B) To make queries slower
- C) To increase storage duplication
- D) To remove all keys

Operating Systems

19. Which page replacement algorithm suffers from Belady's Anomaly?

- A) LRU
- B) FIFO
- C) Optimal
- D) LFU

20. Which of the following is a preemptive scheduling algorithm?

- A) FCFS
- B) Round Robin
- C) Non-preemptive SJF
- D) Batch scheduling

21. What is the purpose of a context switch?

- A) To save the state of a running process and load the state of another
- B) To terminate a process
- C) To allocate memory
- D) To format a disk

22. What is a critical section in OS?

- A) A section of code that accesses shared resources and must not be executed by more than one process/thread at a time
- B) A section reserved for the kernel only
- C) A type of page fault
- D) A memory segment for stack only

23. What is the purpose of a semaphore?

- A) To manage access to shared resources and avoid race conditions
- B) To compile code
- C) To manage file names
- D) To render graphics

24. Which scheduling algorithm gives the shortest average waiting time (in theory, for a given batch)?

- A) FCFS
- B) SJF (Shortest Job First)
- C) Round Robin with a large quantum
- D) Priority Scheduling

25. Which of these describes a 'zombie process'?

- A) A process that has completed execution but still has an entry in the process table
- B) A process that never starts
- C) A process using 100% CPU
- D) A kernel-level thread

26. What does 'thrashing' refer to in an OS?

- A) Excessive CPU usage by a single process
- B) A state where the system spends most of its time swapping pages instead of executing processes
- C) A type of deadlock
- D) A network congestion issue

27. Which of the following is an example of an inter-process communication (IPC) mechanism?

- A) Pipes
- B) Compilers
- C) Loaders
- D) Linkers

28. What is the difference between paging and segmentation?

- A) Paging divides memory into fixed-size blocks; segmentation divides memory into variable-sized logical units
- B) They are identical
- C) Segmentation is faster in all cases
- D) Paging is used only for disk storage

Computer Networks

29. What is NAT (Network Address Translation) used for?

- A) To translate private IP addresses to a public IP address for internet access
- B) To encrypt data
- C) To assign MAC addresses
- D) To resolve domain names

30. Which of the following is a connectionless protocol?

- A) TCP
- B) UDP
- C) FTP
- D) HTTP over TCP

31. What is a firewall used for?

- A) To speed up internet connections
- B) To monitor and control incoming/outgoing network traffic based on security rules
- C) To assign IP addresses
- D) To translate domain names

32. What is the purpose of a subnet?

- A) To divide a large network into smaller, more manageable segments
- B) To increase the number of MAC addresses
- C) To eliminate the need for routers
- D) To replace DNS

Data Structures & Algorithms

33. Which data structure would you use to check if a string of brackets is balanced?

- A) Queue
- B) Stack
- C) Array
- D) Linked List

34. What is the time complexity of Bubble Sort in the worst case?

- A) $O(n)$
- B) $O(n \log n)$
- C) $O(n^2)$
- D) $O(\log n)$

35. What is the time complexity of inserting an element at the head of a singly linked list?

- A) $O(n)$
- B) $O(1)$
- C) $O(\log n)$
- D) $O(n \log n)$

36. What is the time complexity of finding the minimum element in an unsorted array?

- A) $O(1)$
- B) $O(\log n)$
- C) $O(n)$
- D) $O(n \log n)$

37. Which data structure uses FIFO (First In First Out) order?

- A) Stack
- B) Queue
- C) Tree
- D) Graph

38. Which data structure is typically used to implement DFS (iteratively)?

- A) Queue
- B) Stack
- C) Heap
- D) Hash table

39. Which data structure is typically used to implement BFS?

- A) Stack
- B) Queue
- C) Heap
- D) Array only

40. What is a balanced binary search tree used to prevent?

- A) Overflow errors
- B) Degeneration into a linked list (worst-case $O(n)$ operations)
- C) Memory leaks
- D) Integer overflow

41. Which of these best describes the 'sliding window' technique?

- A) Maintaining a subset of elements (a window) over a sequence and adjusting its bounds to solve subarray/substring problems efficiently
- B) A sorting technique for linked lists
- C) A hashing technique
- D) A tree traversal method

42. What is the worst-case time complexity of QuickSort?

- A) $O(n \log n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(n)$

43. Which graph algorithm is used to find the shortest path in a graph with possible negative edge weights (no negative cycles)?

- A) Dijkstra's Algorithm
- B) Bellman-Ford Algorithm
- C) Prim's Algorithm
- D) Kruskal's Algorithm

44. What is dynamic programming primarily used for?

- A) Solving problems by breaking them into overlapping subproblems and storing results to avoid recomputation
- B) Sorting arrays only
- C) Random number generation
- D) Compiling code faster

45. Which of the following is true about a doubly linked list compared to a singly linked list?

- A) It uses less memory per node
- B) Each node has pointers to both the next and previous nodes, allowing traversal in both directions
- C) It cannot be traversed backward
- D) It has no advantages

46. What is the time complexity of accessing an element in an array by index?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n \log n)$

47. Which algorithmic technique does Merge Sort primarily use?

- A) Greedy
- B) Divide and Conquer
- C) Dynamic Programming
- D) Backtracking

48. What is the time complexity of the Two-Pointer technique on a sorted array to find a pair with a given sum?

- A) $O(n^2)$
- B) $O(n \log n)$
- C) $O(n)$
- D) $O(\log n)$

49. What is the key property of a min-heap?

- A) The parent node is always greater than its children
- B) The parent node is always smaller than or equal to its children
- C) It must be a complete binary tree only, with no ordering
- D) It is always a balanced BST

50. What is memoization?

- A) Storing the results of expensive function calls and reusing them when the same inputs occur again
- B) A sorting technique
- C) A type of recursion without a base case
- D) A method to delete duplicate array elements

51. Which of the following problems is a classic example solved using dynamic programming?

- A) Binary search
- B) 0/1 Knapsack problem
- C) Bubble sort
- D) Linear search

52. Which technique explores all possible options and backtracks when a solution path fails, often used in puzzles like N-Queens?

- A) Greedy algorithm
- B) Backtracking
- C) Dynamic programming
- D) Binary search

53. What is the time complexity of binary search on a sorted array of n elements?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n \log n)$
- D) $O(1)$

54. What is the amortized time complexity of appending an element to a dynamic array (like Python's list or Java's ArrayList)?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$ amortized
- D) $O(n^2)$

55. Which of the following is a common technique to resolve hash collisions?

- A) Chaining or open addressing
- B) Binary search
- C) Recursion
- D) Memoization

56. What is the average time complexity of searching in a well-implemented hash table?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(1)$
- D) $O(n^2)$

57. Which data structure is best suited for implementing a priority queue efficiently?

- A) Array
- B) Linked List
- C) Heap
- D) Stack

58. Which data structure uses LIFO (Last In First Out) order?

- A) Queue
- B) Stack
- C) Linked List
- D) Array

59. What is the time complexity of Dijkstra's algorithm using a binary heap?

- A) $O(V^2)$
- B) $O(E \log V)$
- C) $O(V + E)$
- D) $O(E^2)$

60. What is the primary difference between a greedy algorithm and dynamic programming?

- A) Greedy makes a locally optimal choice at each step without reconsidering, DP considers overlapping subproblems and often guarantees a globally optimal solution
- B) They are the same technique
- C) DP never uses recursion
- D) Greedy always gives an optimal solution while DP does not

61. What does BFS stand for in graph traversal?

- A) Best First Search
- B) Breadth First Search
- C) Binary First Search
- D) Backtrack First Search

62. Which sorting algorithm is considered stable and has $O(n \log n)$ time complexity in all cases?

- A) QuickSort
- B) Merge Sort
- C) Selection Sort
- D) Heap Sort

63. What is a monotonic stack typically used for?

- A) Finding the next greater/smaller element efficiently
- B) Sorting an array in place
- C) Implementing recursion
- D) Hashing

64. What is the time complexity of Floyd-Warshall algorithm for all-pairs shortest paths?

- A) $O(V^2)$
- B) $O(V^3)$
- C) $O(E \log V)$
- D) $O(V + E)$

65. Which data structure underlies the implementation of recursion internally?

- A) Queue
- B) Stack (call stack)
- C) Heap (memory heap)
- D) Graph

66. What is tail recursion?

- A) A recursive call that is the last operation in a function, which can be optimized by some compilers into iteration
- B) A recursive function with no base case
- C) A function that calls itself twice
- D) A type of infinite loop

67. Which traversal of a Binary Search Tree gives elements in sorted order?

- A) Pre-order
- B) Post-order
- C) In-order
- D) Level-order

68. What is the space complexity of an iterative approach to computing Fibonacci numbers with memoization storing all n results?

- A) $O(1)$
- B) $O(n)$
- C) $O(n^2)$
- D) $O(\log n)$

69. What is a hash collision?

- A) When two different keys map to the same hash bucket/index
- B) When a hash table runs out of memory
- C) When a key is deleted incorrectly
- D) When two hash tables merge

70. Which algorithm is used to find the Minimum Spanning Tree of a graph?

- A) Dijkstra's Algorithm
- B) Kruskal's or Prim's Algorithm
- C) Bellman-Ford Algorithm
- D) Floyd-Warshall Algorithm

Programming Language Fundamentals

71. What is a memory leak?

- A) When allocated memory is never freed even though it is no longer needed, wasting resources over time
- B) When a program uses too little memory
- C) A type of compilation error
- D) A CPU scheduling issue

72. What does 'immutable' mean for an object (e.g., a String in Java)?

- A) Its value can be changed after creation
- B) Its state cannot be changed after it is created
- C) It can only be used once
- D) It cannot be passed to functions

73. In Python, which of the following is a mutable data type?

- A) Tuple
- B) String
- C) List
- D) int

74. What is the default value of an uninitialized boolean instance variable in Java?

- A) true
- B) false
- C) null
- D) 0

75. What is duck typing (commonly associated with Python)?

- A) A type system that checks types strictly at compile time
- B) An object's suitability is determined by the presence of certain methods/behavior rather than its explicit type
- C) A type of exception
- D) A memory allocation strategy

76. What is the difference between a shallow copy and a deep copy of a list containing nested lists?

- A) A shallow copy also copies nested objects recursively, deep copy does not
- B) A shallow copy copies references to nested objects, deep copy recursively copies nested objects too
- C) They behave identically
- D) Deep copy is not possible in most languages

77. Which of the following is true about Python's Global Interpreter Lock (GIL)?

- A) It allows true parallel execution of threads on multiple cores
- B) It ensures only one thread executes Python bytecode at a time in CPython, limiting CPU-bound multithreading performance
- C) It has no impact on multithreading
- D) It is used only in Python 2

78. In C, what does the 'malloc' function do?

- A) Deallocates memory
- B) Allocates a block of memory dynamically
- C) Initializes a variable to zero
- D) Declares a static variable

79. In Java, what is autoboxing?

- A) Automatic conversion between primitive types and their corresponding wrapper classes
- B) A way to box multiple objects together
- C) A garbage collection technique
- D) A type of casting error

80. What is method chaining?

- A) Calling multiple methods on the same object in a single statement by having each method return the object itself
- B) A way to declare multiple classes
- C) A type of exception handling
- D) A memory management technique

81. Which of the following is a value type (not a reference type) in Java?

- A) int
- B) String
- C) Array
- D) Object

82. What is the purpose of a 'static' variable in a class (Java/C++)?

- A) It belongs to the instance and is unique per object
- B) It belongs to the class rather than any instance and is shared across all instances
- C) It can only be used inside loops
- D) It is always private

Web & Cloud Fundamentals

83. What is the purpose of cookies in web applications?

- A) To store small pieces of data on the client side to maintain state/session information
- B) To encrypt HTTPS traffic
- C) To speed up DNS resolution
- D) To compress images

84. What is the difference between IaaS, PaaS, and SaaS?

- A) They are all the same cloud service model
- B) IaaS provides raw infrastructure, PaaS provides a platform to build apps, SaaS delivers ready-to-use software
- C) SaaS provides infrastructure while IaaS provides software
- D) PaaS is only for storage

85. Which HTTP method is typically used to retrieve data without side effects?

- A) POST
- B) GET
- C) DELETE
- D) PUT

86. What does 'statelessness' mean in the context of REST APIs?

- A) The server stores client session state between requests
- B) Each request from a client contains all the information needed to process it, with no reliance on stored server-side session state
- C) The API has no state at all, including no data
- D) Stateless APIs cannot use authentication

87. What does CSS stand for?

- A) Cascading Style Sheets
- B) Computer Style Sheets
- C) Creative Style System
- D) Cascading Software Sheets

88. What is the purpose of a Content Delivery Network (CDN)?

- A) To cache and serve content from servers geographically closer to users, reducing latency
- B) To encrypt all web traffic
- C) To replace DNS servers
- D) To manage database transactions

89. What does HTTP stand for?

- A) HyperText Transfer Protocol
- B) HighText Transmission Protocol
- C) Hyperlink Transfer Text Protocol
- D) HyperText Trade Protocol

90. What is auto-scaling in cloud computing?

- A) Automatically adjusting compute resources up or down based on demand
- B) Manually adding servers only
- C) A type of database indexing
- D) A billing model only

System Design Basics

91. What is the purpose of a message queue (e.g., Kafka, RabbitMQ) in system design?

- A) To decouple producers and consumers of data, enabling asynchronous processing and buffering
- B) To replace a database
- C) To render a webpage
- D) To manage user sessions only

92. What is database sharding used for?

- A) To horizontally partition a large database across multiple servers to improve scalability
- B) To encrypt a database
- C) To back up data
- D) To normalize tables

93. What is horizontal scaling?

- A) Adding more power (CPU/RAM) to an existing server
- B) Adding more servers/machines to distribute the load
- C) Reducing the number of servers
- D) Only applicable to databases

94. What is eventual consistency?

- A) A guarantee that all replicas are updated instantly
- B) A consistency model where, given enough time without new updates, all replicas converge to the same value
- C) A model where data is never consistent
- D) A synonym for strong consistency

95. What is a load balancer used for in system design?

- A) To store data persistently
- B) To distribute incoming requests across multiple servers to avoid overload on any single server
- C) To compress images
- D) To manage user authentication only

96. What is the purpose of an API Gateway in a microservices architecture?

- A) To act as a single entry point that routes requests to appropriate backend services and can handle cross-cutting concerns like auth and rate limiting
- B) To store all application data
- C) To replace load balancers entirely
- D) To compile microservices code

97. What is the CAP theorem in distributed systems?

- A) A system can simultaneously guarantee Consistency, Availability, and Partition tolerance
- B) A distributed system can only guarantee two of Consistency, Availability, and Partition tolerance at the same time during a network partition
- C) A theorem about compiler design
- D) A theorem unrelated to distributed systems

98. What is idempotency in the context of API design?

- A) An operation that produces the same result no matter how many times it is performed
- B) An operation that can never be repeated
- C) An operation that always fails on retry
- D) A synonym for authentication

99. What is vertical scaling?

- A) Adding more machines to a system
- B) Increasing the resources (CPU/RAM) of an existing machine
- C) Distributing data across regions
- D) A type of caching

100. What is caching used for in system design?

- A) To permanently store all application data
- B) To temporarily store frequently accessed data closer to the application for faster retrieval
- C) To replace a database entirely
- D) To handle user authentication

Answer Key

1-C	2-C	3-A	4-B	5-B	6-B	7-A	8-A	9-B	10-C
11-A	12-B	13-A	14-A	15-B	16-A	17-B	18-A	19-B	20-B
21-A	22-A	23-A	24-B	25-A	26-B	27-A	28-A	29-A	30-B
31-B	32-A	33-B	34-C	35-B	36-C	37-B	38-B	39-B	40-B
41-A	42-B	43-B	44-A	45-B	46-C	47-B	48-C	49-B	50-A
51-B	52-B	53-B	54-C	55-A	56-C	57-C	58-B	59-B	60-A
61-B	62-B	63-A	64-B	65-B	66-A	67-C	68-B	69-A	70-B
71-A	72-B	73-C	74-B	75-B	76-B	77-B	78-B	79-A	80-A
81-A	82-B	83-A	84-B	85-B	86-B	87-A	88-A	89-A	90-A
91-A	92-A	93-B	94-B	95-B	96-A	97-B	98-A	99-B	100-B

Format: [Question No.]-[Correct Option Letter]. Content compiled from publicly available 2026 placement-prep resources and established CS-fundamentals question banks - verify current-year specifics before your interview.